Causal State-Dependent Local Projections

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Motivation: a widely used tool

- An aggregate shock X_t affects micro outcomes $Y_{i,t+h}$.
- A predetermined state $Z_{i,t-1}$ describes financial conditions or household characteristics.

$$\text{Scalar LP:} \quad Y_{i,t+h} = \beta_h X_t + \cdots$$
$$\text{Linear state dependence:} \quad Y_{i,t+h} = (\alpha_h + \beta_h Z_{i,t-1}) X_t + \cdots$$

When does an LP recover the causal response as a function of the state?

Gap in the literature

- Scalar LPs can identify averages of causal effects under the relevant exogeneity conditions, even in nonlinear DGPs.
- Recent work gives interaction coefficients weighted-average or projection interpretations.
- A causal average or projection need not recover the response function over the state.
- Endogenous states can coexist with identification. Nonlinear shock effects and shock-dependent propagation are central obstacles.

Kolesár–Plagborg-Møller (2024), Gonçalves et al. (2024), Almuzara–Sancibrián (2025), Winkler (2026), as discussed in the manuscript.

This paper

1. **Identification**
   A conditional mean linear in the shock, together with causal exogeneity and an m.d.s. shock, identifies a causal response function.

2. **Specification**
   Linear interactions generally recover a projection. Flexible state dependence recovers the function under the identifying conditions.

3. **Inference**
   Sieve estimation with pointwise and state-uniform inference accommodates convergence rates that vary over the state space.

Causal IRF: the object of interest

Let $Y_{i,t+h}(x)$ denote the outcome when the date-t shock is set to x.

$$\text{IRF}_h(z;\delta) = E[Y_{i,t+h}(X_t + \delta) - Y_{i,t+h}(X_t) \mid Z_{i,t-1} = z].$$

- Condition on the **initial** state. Later states may respond endogenously.
- Change one date-t shock, allowing subsequent dynamics to unfold.
- Maintain consistency and admissible conditional support for the finite shift.

Target representation: $\text{IRF}_h(z;\delta) = \delta g_h(z)$.

Three stylised DGPs

DGP 1: linear shock exposure and fixed propagation

$$Y_{it} = a(Z_{i,t-1})X_t + \rho Y_{i,t-1} + \varepsilon_{it}. \qquad g_h(z) = \rho^h a(z)$$

DGP 2: shock-dependent propagation

$$Y_{it} = a(Z_{i,t-1})X_t + \rho_t Y_{i,t-1} + \varepsilon_{it}.$$

The date-t shock affects future ρ_{t+j}, so the response generally loses the multiplicative form.

DGP 3: nonlinear dependence on the shock

$$Y_{it} = f(Z_{i,t-1}, X_t) + \rho Y_{i,t-1} + \varepsilon_{it}.$$

The finite response generally depends on both X_t and δ.

These statements maintain the paper's shock and innovation restrictions.

Summary: what LPs recover

	DGP 1	DGP 2	DGP 3
Scalar LP	Average causal effect	Average causal effect	Average causal effect
Linear state dependence	Projection; IRF if g_h is linear	Projection	Projection
Flexible state dependence	Causal $g_h(z)$	Projection	Projection

- Each causal claim maintains the relevant exogeneity conditions.
- Flexibility in the state does not repair nonlinear shock effects or shock-dependent propagation.

Identification: the key condition

A1: causal mean exogeneity and an m.d.s. shock

$$E[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}, X_t] = E[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}], \quad E[X_t \mid \mathcal{F}_{t-1}] = 0.$$

A2: $Z_{i,t-1}$ and controls are $\mathcal{F}_{t-1}$-measurable.

A3: horizon-specific conditional mean

$$E[Y_{i,t+h} \mid \mathcal{F}_{t-1}, X_t] = g_h(Z_{i,t-1})X_t + r_{i,h}(\mathcal{F}_{t-1}).$$

Conclusions, with consistency, support and positive shock variance:

$$\mathrm{IRF}_h(z;\delta) = \delta g_h(z), \qquad g_h(z) = \frac{E[X_t Y_{i,t+h} \mid Z_{i,t-1} = z]}{E[X_t^2 \mid Z_{i,t-1} = z]}.$$

The unrestricted nuisance $r_{i,h}$ need not be estimated.

Where A3 holds (I): fixed types and HANK

- Fixed characteristics can index heterogeneous exposure. A fixed state **alone does not imply A3**.
- Within a first-order HA/HANK approximation, fixed linear propagation and linear aggregate-shock exposure can yield

$$E[Y_{i,t+h} \mid \mathcal{F}_{t-1}, X_t] = g_h(s_{i,t-1})X_t + r_{i,h}(\mathcal{F}_{t-1}).$$

- The loading can vary nonlinearly across micro states, reflecting MPCs and balance-sheet exposure.
- For the underlying nonlinear economy, the approximation must remain accurate over the shock range considered.

Where A3 holds (II): macro-finance panel VAR

$$V_{it} = A_i(L)V_{i,t-1} + \lambda(s_{i,t-1})X_t + e_{it}, \qquad V_{it} = (Y_{it}, W_{it})^\top.$$

- Time-invariant propagation may **differ across units**.
- Exposure $\lambda(\cdot)$ can be nonlinear in the predetermined micro state.
- Under the innovation restrictions, iteration collects direct and indirect channels in the horizon-h response.
- The chosen state must capture the relevant response heterogeneity for the exact A3 representation. Otherwise Proposition 1 gives a weighted subpopulation response.

Deterministic/predictable additive terms are suppressed here. Propagation heterogeneity can affect g_h itself.

Where A3 fails

- **Nonlinear shock effects**
  Sign asymmetries, large discrete interventions, regime switching and occasionally binding constraints.
- **Shock-dependent propagation**
  The shock changes the persistence or riskiness of later dynamics.
- **Fully nonlinear HA/HANK environments**
  The induced path can move across regions with different local dynamics.

Almuzara et al. (2025), as discussed in the paper, model household-income exposure and persistence varying with aggregate conditions.

Does the choice of state matter?

If the true response depends on a richer predetermined state s, the functional LP based on Z identifies

$$g_h^{LP}(z) = E[w(s;z)\widetilde{g}_h(s) \mid Z = z], \qquad w(s;z) = \frac{E[X_t^2 \mid s, Z = z]}{E[X_t^2 \mid Z = z]}.$$

- $w(s;z) \geq 0$ and $E[w(s;z) \mid Z = z] = 1$.
- This is a **variance-weighted causal response** within the observed-state group.
- The ordinary group-average finite response is $\delta E[\widetilde{g}_h(s) \mid Z = z]$.
- The two coincide under additional conditions, such as shock variance constant across latent types within the group.

Why standard linear LPs miss the function

Proposition 2. The coefficient on $Z_{i,t-1}X_t$ averages local slopes of the causal response:

$$\beta_h = \int_{\mathcal{Z}} \omega_h(z) g_h'(z)\, dz, \qquad \omega_h(z) \geq 0, \quad \int_{\mathcal{Z}} \omega_h(z)\, dz = 1.$$

- Nonnegative weights preserve an average-derivative interpretation.
- A positive average slope does not establish monotonicity.
- The fitted line generally misses $g_h(z)$ and its state contrasts.

Example: a zero slope can hide substantial heterogeneity

$$Z \sim U[-1,1], \quad X \perp Z, \quad g(z) = 1 - z^2.$$

$$\beta = 0, \qquad g(0) = 1, \qquad g(\pm 1) = 0.$$

The regression includes X and ZX and maintains the shock restrictions. The fitted response is the constant $2/3$. Weight derivation: backup slide 32.

Proposed solution: sieve LP

$$Y_{i,t+h} = X_t \phi_J(Z_{i,t-1})^\top b_h + W_{i,t-1}^\top \gamma_h + u_{i,t+h}.$$

$$\widehat{g}_h(z) = \phi_J(z)^\top \widehat{b}_h.$$

- Cubic B-splines with a constant in their span and knots at empirical quantiles.
- Estimate the shock-basis interactions and controls by OLS.
- Choose $J \in \{4, \ldots, 20\}$ by AIC in the implementation.
- $J = 4$ spans a global cubic polynomial. A linear basis is a separate restricted specification.
- Formal theory uses a growing sequence of J satisfying its bias and rate conditions.

Inference: state-dependent convergence rates

Let $\Omega_{h,J,N}$ be the long-run covariance of the $N^{-1/2}$-normalized effective score, and

$$\chi_{h,J,N} = 1 \vee \lambda_{\max}(\Omega_{h,J,N}).$$

Theorem 1: coefficient rate

$$\|\widehat{b}_h - b_h\|_2 = O_p(J^{1/2-\kappa}) + O_p\left(\sqrt{\frac{J\chi_{h,J,N}}{NT}}\right).$$

- Bounded χ: stochastic coefficient error $\sqrt{J/(NT)}$.
- $\chi \asymp N$: stochastic coefficient error $\sqrt{J/T}$.
- Aggregate uncertainty can load differently at different states, so pointwise rates may vary with z.

Assumption A.1 and the stated joint growth restrictions apply. The pointwise CLT additionally requires state-specific undersmoothing, linearization and feasible covariance consistency.

Inference: pointwise and uniform theory

Pointwise inference: under the additional conditions of Theorem 1,

$$\frac{\widehat{g}_h(z) - g_h(z)}{\widehat{\sigma}_h(z)} \Rightarrow N(0,1) \quad \text{for each fixed } z.$$

Theorem 2: under the stronger uniform conditions,

$$\sup_{z \in \mathcal{Z}} |\widehat{g}_h(z) - g_h(z)| = O_p(J^{1-\kappa}) + O_p\left(J\sqrt{\frac{\chi_{h,J,N}}{NT}}\right).$$

Feasible covariance-kernel consistency additionally yields

$$P\{g_h(z) \in C_h(z) \text{ for every } z \in \mathcal{Z}\} \longrightarrow 1 - \alpha.$$

- Horizon h is fixed. The band is uniform over states.
- Bias, dependence, sieve growth and Gaussian approximation conditions are essential.
- AIC selection uncertainty is outside the formal theorem.

Inference: feasible HAC covariance

Partial out controls from $P_{it} = X_t \phi_J(Z_{i,t-1})$:

$$\widetilde{P}_{h,it} = P_{it} - \widehat{A}_{12,h}\widehat{A}_{22,h}^{-1} W_{i,t-1}, \qquad \widehat{s}_{h,t} = \frac{1}{\sqrt{N}} \sum_i \widetilde{P}_{h,it} \widehat{u}_{i,t+h}.$$

Estimate its long-run covariance using Bartlett HAC:

$$\widehat{\Omega}_h = \widehat{\Gamma}_h(0) + \sum_{k=1}^{L_h} \left(1 - \frac{k}{L_h + 1}\right) \{\widehat{\Gamma}_h(k) + \widehat{\Gamma}_h(k)^\top\}.$$

With $\widehat{\widetilde{A}}_{11,h}$ the residualized Gram matrix,

$$\widehat{V}_h = \frac{\widehat{\widetilde{A}}_{11,h}^{-1} \widehat{\Omega}_h \widehat{\widetilde{A}}_{11,h}^{-1}}{N(T-h)}, \qquad \widehat{\sigma}_h^2(z) = \phi_J(z)^\top \widehat{V}_h \phi_J(z).$$

$\widehat{\Gamma}_h(k) = (T-h)^{-1} \sum_{t=k+1}^{T-h} \widehat{s}_{h,t} \widehat{s}_{h,t-k}^\top$. The paper uses $L_h = \lfloor 4[(T-h)/100]^{2/9} \rfloor$ and checks $L_h = h$.

Inference: a studentized uniform band

At a fixed horizon, simulate $\xi^{(b)} \sim N(0, I_J)$ and set

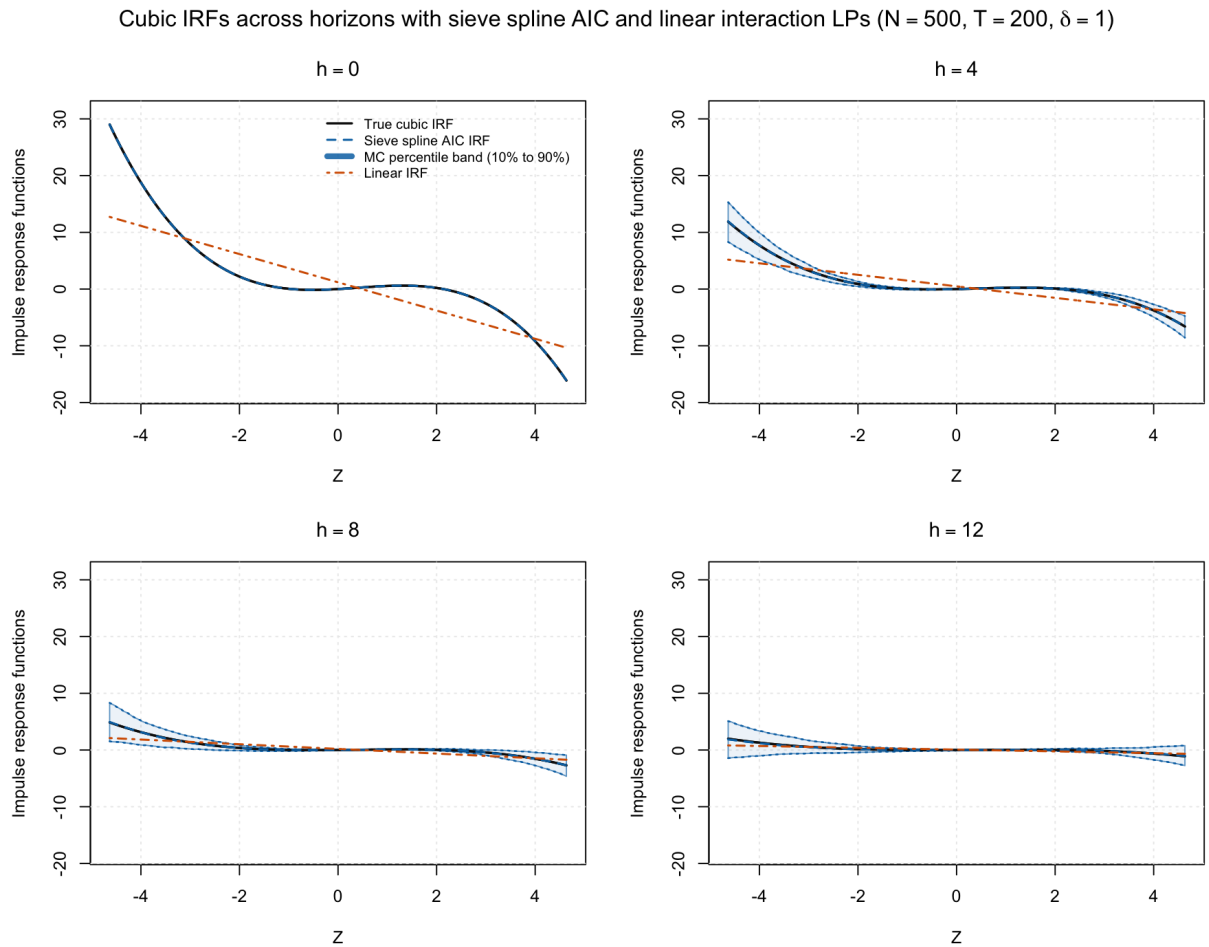
$$G_h^{(b)}(z) = \phi_J(z)^\top \widehat{V}_h^{1/2} \xi^{(b)}, \qquad M_h^{(b)} = \max_{z \in \mathcal{Z}_G} \frac{|G_h^{(b)}(z)|}{\widehat{\sigma}_h(z)}.$$

Let $c_{h,1-\alpha}^{stu}$ be the empirical $(1-\alpha)$ quantile of the maxima.

$$C_h(z) = \left[\widehat{g}_h(z) \pm c_{h,1-\alpha}^{stu} \widehat{\sigma}_h(z)\right].$$

- Studentize **before** taking the maximum.
- The two-sided band uses the absolute value.
- The grid approximates the continuum result under the theorem's conditions.
- Uniform bands quantify function uncertainty. Shape claims require an appropriate comparison or test.

Simulation: sieve vs. linear across horizons



$g(z) = 0.5z + 0.3z^2 - 0.25z^3$, $g_h(z) = 0.8^h g(z)$, $N = 500$, $T = 200$. Figure 2: Monte Carlo averages and 10–90% simulation percentiles, **not confidence bands**.

Intermediate shocks and precision

AIC, $N = 500$, $T = 200$, nominal 95% state-uniform bands:

	Without intermediate terms		With intermediate terms	
h	Coverage	Width	Coverage	Width
0	0.86	0.03	0.86	0.03
4	0.89	1.30	0.83	0.05

- In this DGP, omitted future-shock terms leave common variation in LP errors without changing identification.
- Including the terms narrows bands, but does not guarantee improved coverage.
- Added interactions increase dimension and can cause instability. The paper retains the unaugmented baseline.

Selector comparison

At $h = 4$, $(N, T) = (500, 200)$, nominal 95%:

Selector	Coverage	Average width
Oracle ($J = 4$)	0.92	1.17
AIC	0.89	1.30
GCV	0.89	1.30
LASSO	0.90	1.42

- Data-driven selectors give similar results in this design.
- All entries remain below nominal coverage, including the oracle.
- Coverage generally improves with T. Short-panel undercoverage remains a limitation.

Empirical application: firm investment

$$k_{i,t+h} - k_{i,t-1} = \alpha_{sth} + g_h(Z_{i,t-1})X_t + \rho_h \Delta k_{i,t-1} + W_{i,t-1}^{\top}\gamma_h + u_{i,t+h}.$$

- Compustat quarterly capital stock, with cumulative net investment as the outcome.
- High-frequency monetary surprises. Figures show a **25 b.p. surprise rate cut**.
- Standardized distance to default (D2D): higher values mean safer firms.
- Sector-time-horizon fixed effects absorb the common response. g_h needs a level normalization, while state contrasts are invariant.
- Relative to OW: lagged investment, the treatment of financial-position demeaning and firm fixed effects also differ.

Firm investment IRFs: linear vs. nonlinear

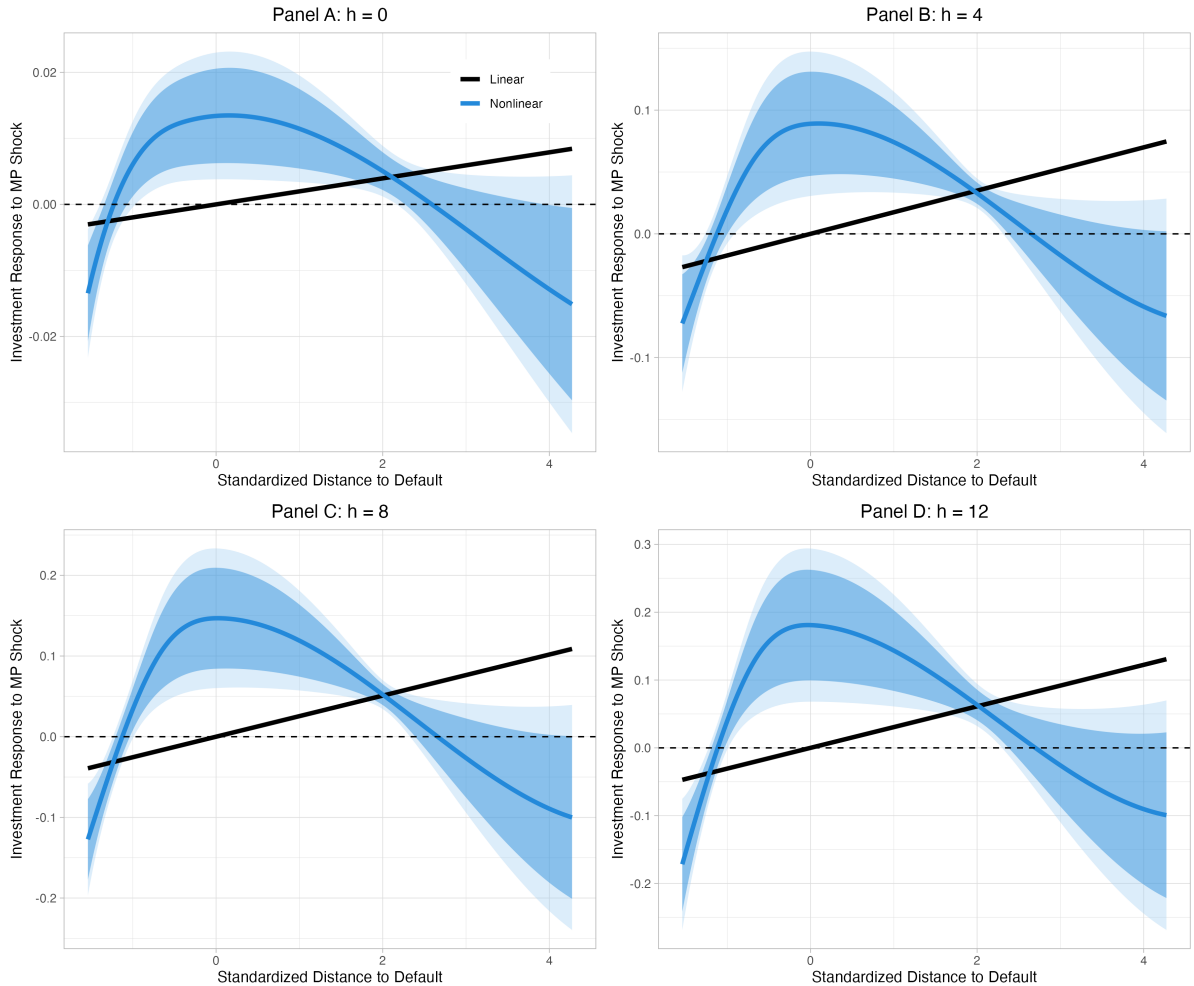
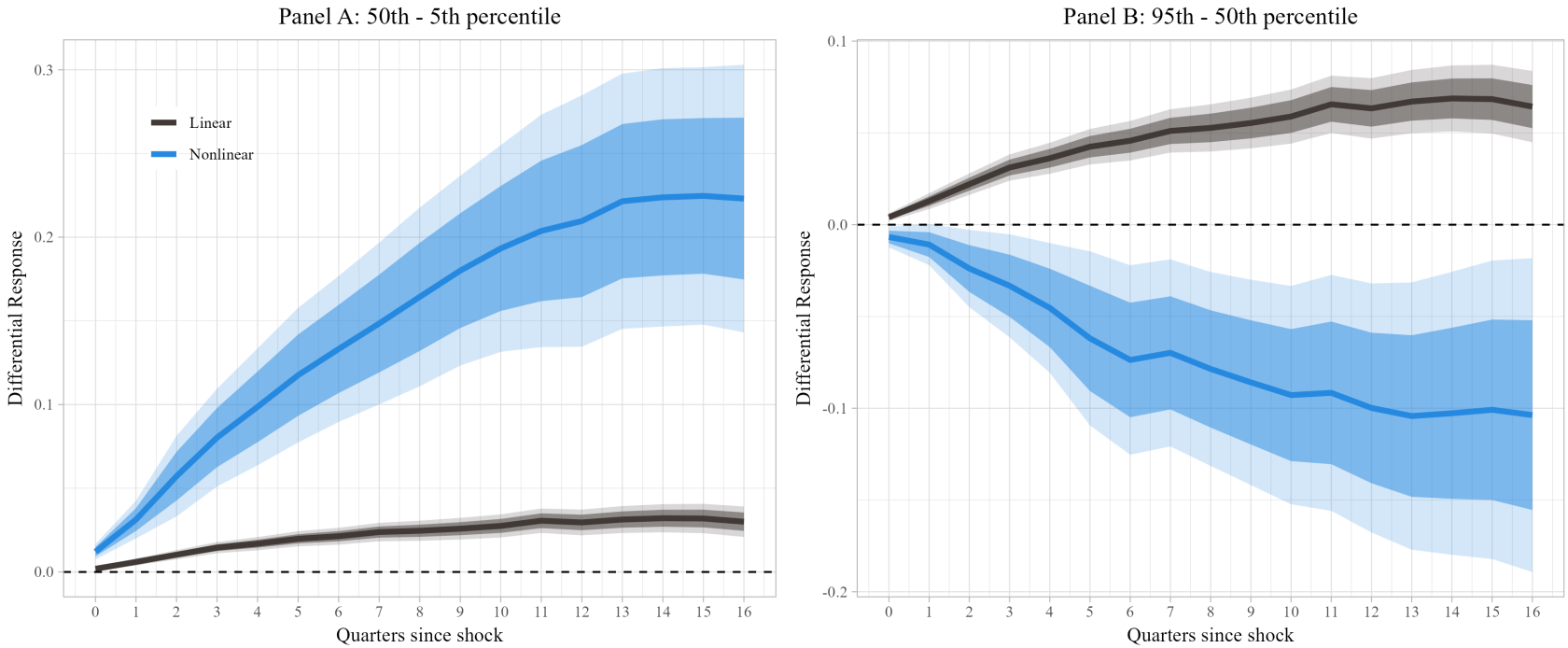


Figure 3. Differential cumulative responses to a 25 b.p. rate cut. Horizons 0, 4, 8 and 12 quarters. Dark/light shading: 68%/90% state-uniform bands at each horizon.

Firm investment IRFs: interpretation

- The flexible response peaks near mean D2D and falls toward both tails.
- In the **left tail**, responsiveness increases as firms become safer.
- To the **right of the peak**, responsiveness decreases as firms become safer.
- A possible mechanism is limited responsiveness near default and strong interest sensitivity in the middle. The estimates do not identify that mechanism uniquely.
- The linear slope remains a projection summary. It does not establish monotonicity of the full response.

Differential responses across D2D



Panel A: 50th minus 5th percentile. Panel B: 95th minus 50th percentile.

$$\widehat{\Delta}_h = \{\phi_J(z_a) - \phi_J(z_b)\}^\top \widehat{b}_h, \quad \widehat{\mathrm{Var}}(\widehat{\Delta}_h) = a^\top \widehat{V}_h a, \quad a = \phi_J(z_a) - \phi_J(z_b).$$

Figure 4 uses 68% and 90% pointwise intervals for each fixed-state contrast at each horizon.

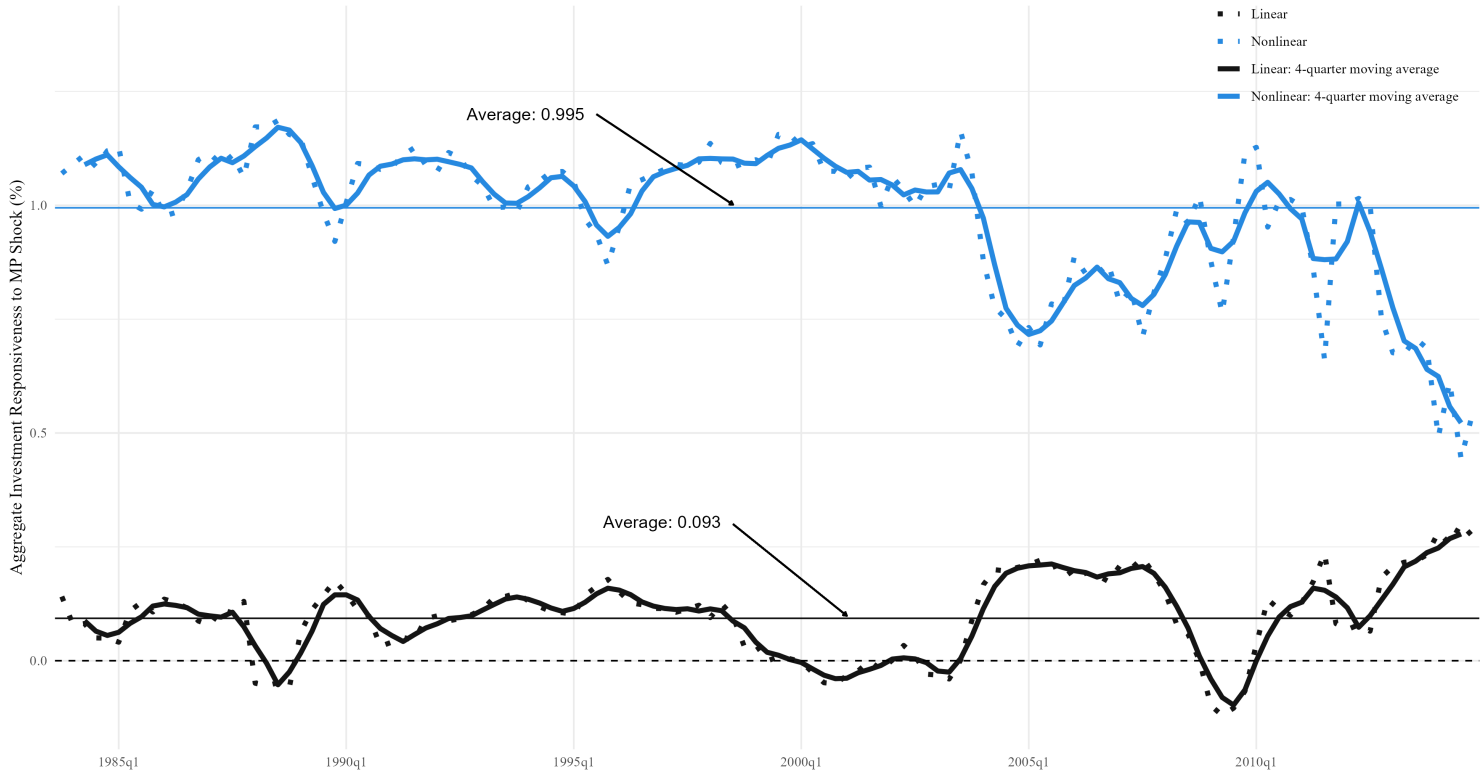
Differential responses: takeaways

At 16 quarters, for a 25 b.p. rate cut:

Contrast	Linear	Nonlinear
50th minus 5th percentile	Just under 5%	Above 20%
95th minus 50th percentile	Almost +10%	About −10%

- The linear specification understates the left-tail contrast.
- The right-tail comparison reverses sign.
- These are approximate magnitudes reported in the manuscript, expressed using its log-response percentage convention.

Aggregate implications



Capital-weighted **heterogeneity component**, relative to the sector-time benchmark:

$$\delta \sum_i \frac{K_{i,t-1}}{\sum_j K_{j,t-1}} \widehat{g}_h(Z_{i,t-1}).$$

Figure 5 reports impact responsiveness. The common aggregate response is absorbed by fixed effects and is not recovered by this calculation.

Aggregate implications: takeaways

- Mean impact contribution of financial heterogeneity:
  Nonlinear: about 1.0 p.p. **Linear: about 0.09 p.p.**
- Figure 5 labels the averages 0.995 and 0.093. The contribution is roughly an order of magnitude larger with flexible state dependence.
- The paper reports a correlation of about -0.7 between the two time profiles.
- Moving mass toward the high-response middle can raise this component even as average financial health deteriorates.

All comparisons retain the paper's normalization. They do not measure a tenfold increase in the total monetary-policy effect.

Conclusion

1. Causal shock exogeneity needs a conditional-mean structure to identify the response **function**.

2. Flexible state dependence recovers that function when the identifying conditions hold. A linear interaction generally summarizes average slopes.

3. Common shocks make the time dimension and state-specific uncertainty central to inference.

4. Firm investment responses are hump-shaped in financial conditions, with consequential heterogeneity and aggregation differences.

Thank you

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Appendix: identification proof sketch

Causal step: by consistency, A1(i) and A3,

$$E[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}] = g_h(Z_{i,t-1})x + r_{i,h}(\mathcal{F}_{t-1}).$$

Subtract and condition on $Z_{i,t-1} = z$ to obtain $\mathrm{IRF}_h(z;\delta) = \delta g_h(z)$.

Moment step: A1(ii) removes the nuisance,

$$E[X_t r_{i,h}(\mathcal{F}_{t-1}) \mid Z] = E[r_{i,h}(\mathcal{F}_{t-1})E[X_t \mid \mathcal{F}_{t-1}] \mid Z] = 0.$$

Therefore

$$E[X_t Y_{i,t+h} \mid Z = z] = g_h(z)E[X_t^2 \mid Z = z].$$

Finite shifts require conditional support. Division requires positive conditional shock variance.

Appendix: Proposition 2 proof idea

Write $X = X_t$, $Z = Z_{i,t-1}$ and $\mu_{Z,X} = E[X^2 Z]/E[X^2]$. Residualizing ZX on X gives $X(Z - \mu_{Z,X})$. Hence

$$\beta_h = \frac{E[X^2(Z - \mu_{Z,X})g_h(Z)]}{D}, \qquad D = E[X^2(Z - \mu_{Z,X})^2].$$

Use $g_h(Z) = g_h(\underline{z}) + \int_{\underline{z}}^{\overline{z}} 1\{Z \geq u\} g_h'(u)\, du$:

$$\beta_h = \int \frac{E[X^2(Z - \mu_{Z,X})1\{Z \geq u\}]}{D} g_h'(u)\, du.$$

- The constant term cancels because $E[X^2(Z - \mu_{Z,X})] = 0$.
- Upper-tail centered moments are nonnegative under the X^2-weighted distribution.
- Integrating the numerator over u gives D, so weights integrate to one.

Appendix: numerical illustrations

A zero interaction can hide a hump

$$Z \sim U[-1,1], \quad X \perp \mathcal{F}_{t-1}, \quad E[X] = 0, \quad g(z) = 1 - z^2.$$

$$\beta = \frac{\mathrm{Cov}(Z, 1 - Z^2)}{\mathrm{Var}(Z)} = 0, \quad \alpha = \tfrac{2}{3}, \quad g(0) = 1, \quad g(\pm 1) = 0.$$

Coarsening can change population weights

Two equally frequent latent types: causal slopes 1 and 3, conditional shock variances 1 and 4, and zero conditional shock means.

$$\text{Group-average slope} = 2, \qquad \text{LP slope} = \frac{0.5(1)(1) + 0.5(4)(3)}{0.5(1) + 0.5(4)} = 2.6.$$

Both examples are analytical illustrations, not estimates from the application.

Appendix: impact coverage with AIC

Nominal 95%, $h = 0$. Current main-text results:

	$N = 100$		$N = 500$	
T	Coverage	Width	Coverage	Width
40	0.73	0.14	0.72	0.06
120	0.83	0.08	0.84	0.04
200	0.88	0.06	0.86	0.03

- 300 replications, 2,000 Gaussian draws, 500 state-grid points.
- Coverage is evaluated over the 5th–95th percentile state range.
- Larger T improves coverage, but the reported values remain below 0.95.

Appendix: coverage and RIMSE at later horizons

AIC at $h = 4$, $N = 500$, nominal 95%

	Without intermediate terms		With intermediate terms	
T	Coverage	Width	Coverage	Width
40	0.75	2.56	0.67	0.10
120	0.87	1.64	0.81	0.06
200	0.89	1.30	0.83	0.05

RIMSE for a unit shock, $N = 500$, $T = 200$

h	0	4	8	12
Sieve	0.009	0.836	0.804	0.795
Linear	4.506	1.962	0.997	0.683

Appendix: economic interpretation of A3

- The date-t shock enters each horizon's conditional mean linearly.
- The coefficient captures direct exposure and indirect propagation effects.
- Time-invariant propagation may be heterogeneous across units. This heterogeneity can affect the response coefficient.
- An observed state that omits relevant heterogeneity gives the weighted interpretation in Proposition 1.
- Endogenous predetermined states are admissible. Shock-dependent changes in the propagation law generally are not.

Appendix: HA/HANK in more detail

Within a first-order aggregate-perturbation representation,

$$q_t = Aq_{t-1} + BX_t, \qquad \Delta Y_{it} = C_i q_t,$$

with fixed propagation and linearized outcome mapping,

$$\frac{\partial \Delta Y_{i,t+h}}{\partial X_t} = C_i A^h B.$$

- C_i can vary nonlinearly across predetermined micro states.
- For an underlying nonlinear outcome map, its first-order expansion must also be used.
- Finite-shock interpretation in the nonlinear economy requires an accurate approximation over the compared shock range.
- Coarsening yields variance weights unless additional conditions remove them.

Appendix: macro-finance VAR in more detail

For a one-lag illustration with fixed A_i,

$$V_{it} = A_i V_{i,t-1} + \lambda(s_{i,t-1})X_t + e_{it}.$$

Under the relevant future-innovation mean restrictions, the contribution of the date-t shock to $Y_{i,t+h}$ is

$$e_Y^\top A_i^h \lambda(s_{i,t-1})X_t.$$

- Both exposure and propagation enter the horizon coefficient.
- If the observed state summarizes both, this is $g_h(Z_{i,t-1})X_t$.
- Otherwise interpret the functional LP via Proposition 1.
- A shock-dependent A_i generally invalidates this fixed-power derivation.

Appendix: measured shocks and controls

If a measured series M_t has a predictable component, write

$$M_t = m(\mathcal{F}_{t-1}) + X_t, \qquad E[X_t \mid \mathcal{F}_{t-1}] = 0.$$

- The identifying restrictions must hold for the innovation X_t used in the LP, including the causal mean restriction.
- Residualizing M_t addresses predictability. It does not by itself establish causal exogeneity.
- Predetermined controls may improve precision. A short linear lag list need not remove an arbitrary predictable component.
- The paper treats the shock as observed. Estimated residuals or an IV first stage require additional inference arguments.

The last point includes generated-shock uncertainty and weak-instrument issues, which the paper's stated inference theory abstracts from.

Appendix: fixed effects and normalization

With sector-time-horizon fixed effects, for any constant c_h,

$$g_h^*(z) = g_h(z) + c_h, \qquad \alpha_{sth}^* = \alpha_{sth} - c_h X_t$$

leaves fitted values unchanged.

- The shock-only regressor is collinear with these fixed effects. A level restriction is required for estimation.
- Contrasts are identified independently of that level:

$$g_h^*(z_a) - g_h^*(z_b) = g_h(z_a) - g_h(z_b).$$

- A capital-weighted average changes by δc_h under this shift, since capital shares sum to one.
- The reported aggregate calculation therefore retains the paper's benchmark. It is the heterogeneity component relative to that benchmark.